

Supplementary material to the manuscript Edwards Localization

Riccardo Fantoni*

Università di Trieste, Dipartimento di Fisica, strada Costiera 11, 34151 Grignano (Trieste), Italy

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We give the listings of the Wolfram Mathematica code used to generate Fig. 1 and 2.

Keywords: Polaron localization; Anderson localization; Edwards localization

I. THE CODE

We generated the data of Fig. 1 & 3 with the following Wolfram Mathematica [1] code. Note that the phases φ_i are here calculated analytically, with the command `Solve`, and as a consequence the code becomes extremely slow for a number of scattering centers $N > 3$. In order to study the thermodynamic limit where it is necessary to increase N at constant density $\rho = N/L$ it is necessary to calculate the $N + 1$ phases numerically using `FindRoot`.

```
(*****  
(* Mathematica Wolfram Code for Fig. 1 & 2 *)  
*****)  
  
(* wave function between the scattering centers *)  
Psi[x_, k_, A_, phi_] := A*Sin[k*x + phi];  
dPsi[x_, k_, A_, phi_] := A*k*Cos[k*x + phi];  
  
(* normalization condition for the wave function *)  
AAA[k_, phi_] :=  
  A /. Solve[Integrate[Psi[x, k, A, phi]^2, {x, 0, L}] == 1, A][[1]]  
  
(* transfer matrices *)  
RR[theta_] := {{Cos[theta], -Sin[theta]}, {Sin[theta], Cos[theta]}};  
TT[m_, g_, k_] := {{1, 2*m*g/k}, {0, 1}};  
  
(* PARAMETERS *)  
Clear[AA, phi, xx]  
NN = 3; (* number of scattering sites NN=1,2,3,... *)  
L = 1; (* segment length [0,L] *)  
m = 1; (* mass of the particle *)  
  
(* RECURRENCE RELATION FOR AMPLITUDES *)  
Clear[AA];  
Do[AA[i + 1] =  
  AA[i + 1] /.  
    Solve[dPsi[xx[i + 1], k, AA[i + 1], phi[i + 1]] ==  
      dPsi[xx[i + 1], k, AA[i], phi[i]] +  
        2*m*g*Psi[xx[i + 1], k, AA[i], phi[i]], AA[i + 1]][[1]], {i, 0,  
      NN - 1}];  
  
(* RECURRENCE RELATION FOR PHASES *)  
Clear[phi];  
Do[phi[i + 1] =  
  phi[i + 1] /.  
    Solve[Psi[xx[i + 1], k, AA[i + 1], phi[i + 1]] ==  
      Psi[xx[i + 1], k, AA[i], phi[i]], phi[i + 1]][[1]], {i, 0,  
      NN - 1}];  
  
(* NORMALIZATION OF EIGENSTATE (amplitude AA[0]) *)  
Do[AA[i] = AAA[k, phi[i]], {i, 0, 0}];  
  
(* MONTE CARLO *)  
M = 100; (* MC Steps *)  
g = 100; (* coupling constant between particle and scattering centers *)  
tt = 1; (* time *)  
  
SeedRandom[1];  
PsiMC[x_] = 0;  
PsiMCT[x_] = 0;  
EMC = 0;  
Do[
```

```
(* RANDOM POSITIONS OF SCATTERING CENTERS *)  
Clear[xxxx, xx];  
Do[xxxx[i] = RandomReal[]*L, {i, 1, NN}];  
xx[0] = 0;  
  
Do[xx[i] = Sort[Array[xxxx, NN]][[i]], {i, 1, NN}];  
Do[Print["r(", i, ") = ", xx[i]], {i, 1, NN}];  
Print["-----1-----"];  
(* EIGENVALUE (search for the ground state with Newton-Raphson) *)  
Clear[MM, k];  
MM[0] = {{1, 0}, {0, 1}};  
Do[MM[i + 1] =  
  Dot[Dot[RR[k*(xx[i + 1] - xx[i])], TT[m, g, k]], MM[0]], {i, 0,  
    NN - 1}];  
Print["-----2-----"];  
FR = FindRoot[Tr[MM[NN]] == 1, {k, 0.01}];  
k = k /. FR;  
EE = k^2/2/m;  
Print["E(", j, ") = ", EE];  
(* PBC DETERMINATION OF PHI[0] *)  
RRR = FindRoot[  
  Psi[0, k, AA[0], phi[0]] == Psi[L, k, AA[NN], phi[NN]], {phi[0],  
    1}];  
phi[0] = phi[0] /. RRR;  
Print["-----3-----"];  
xx[NN + 1] = L;  
PsiF[x_] =  
  Piecewise[  
    Table[{Psi[x, k, AA[i], phi[i]], xx[i] <= x < xx[i + 1]}, {i, 0,  
      NN}];  
PsiFt[x_] = Exp[-I*EE*tt]*PsiF[x];  
PsiMC[x_] = PsiMC[x] + PsiF[x];  
PsiMCT[x_] = PsiMCT[x] + PsiFt[x];  
EMC = EMC + EE;  
Print["EMC = ", EMC/j];  
, {j, 1, M}]  
  
Plot[PsiMC[x]/M, {x, 0, L}, PlotRange -> Full]  
Plot[Abs[PsiMCT[x]/M]^2, {x, 0, L}, PlotRange -> Full]
```

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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* riccardo.fantoni@scuola.istruzione.it

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- [1] Wolfram Mathematica <https://www.wolfram.com/mathematica/>.